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Mastering Boss Combat in Cuphead: A Reinforcement Learning Approach with Procedural Difficulty Curriculum

Introduction

This project aims to develop an intelligent agent capable of defeating boss fights in Cuphead, a highly challenging 2D platformer (over 6 million sales), using reinforcement learning (RL) for autonomous combat in a fast-paced, pattern-based environment. While previous work has applied supervised learning to Cuphead gameplay, no published research has successfully trained an RL agent to master these complex, multi-phase boss fights from scratch. This project addresses that gap by introducing a custom game-state extraction mod, procedural difficulty curricula to accelerate training, and dynamic attack pattern modelling for adaptive dodging strategies.

Background

Cuphead boss fights present significant challenges for AI: sub-second decision windows, complex multi-phase attack patterns, and precise movement requirements. Bosses cycle through 2–5 distinct phases, each introducing new mechanics and telegraphed, semi-predictable sequences. Standard reactive behaviour trees fail here due to required temporal credit assignment and delayed rewards. The project leverages Proximal Policy Optimization (PPO), which has proven stable in continuous control tasks. To access the environment without ToS violations, the project will utilise the Blender modding API (a BepInEx-based C# framework) to hook into Cuphead's assemblies and extract clean game state, avoiding the computational overhead of pixel-based computer vision.

Proposed Project

The project consists of three integrated research components:

- 1. Game State Extraction:** Develop a BepInEx/Blender C# mod to extract precise state vectors: player position/HP, boss phase/HP, projectile positions/velocities, and parry-able object locations. This formalises the fight as an episodic Markov Decision Process (MDP) connected to a Python-based RL training loop via inter-process communication (IPC).
- 2. Procedural Difficulty Curriculum:** Implement four curriculum training variants: baseline (full difficulty), speed-scaled (projectile speeds ramped from 0.5x to 1.0x), phase-isolated (training on individual boss phases before full fights), and telegraph-extended (lengthened visual warnings). Agents will train on 1–2 Inkwell Isle I bosses, advancing through curriculum thresholds (e.g., 75%-win rate) to measure sample efficiency gains versus the baseline.
- 3. Attack Pattern Modelling:** Because Cuphead bosses follow phase-specific, telegraphed sequences rather than random attacks, an LSTM neural network will be built to predict the next attack based on the sequence history. The RL policy will be conditioned on these predictions to enable pre-emptive positioning and optimise damage windows.

Technical Approach: Continuous control using the PPO algorithm; automated curriculum progression; C# modding (BepInEx/Blender) combined with a Python IPC bridge for the RL training loop.

Timeline

Weeks 1-2: Blender API mod + state extraction

Weeks 3-4: Baseline PPO on single boss

Weeks 5-6: Curriculum system (4 variants)

Weeks 7-9: Curriculum training experiments

Weeks 10-12: LSTM pattern modelling + adaptive policy

Weeks 13-14: Evaluation and ablation studies

Week 15: Research paper (10-15 pages, IEEE/Springer format)

References:

1. Millington, I., & Funge, J. (2025). **Artificial Intelligence and Games** (2nd ed., Chapters 3-5).
 2. Schulman, J., et al. (2017). **Proximal Policy Optimization Algorithms**.
 3. Mnih, V., et al. (2015). Human-level control through deep reinforcement learning
 4. Bengio, Y., et al. (2009). Curriculum Learning. **Proceedings of ICML**
 5. Blender Modding API Documentation. **Cuphead Modding Wiki**
 6. Studio MDHR (2017). **Cuphead**. [Game]. 6M+ sales; 96% positive
 7. **Game Maker's Toolkit** (2017). "How Cuphead's Bosses (Try to) Kill You."
- Note:** 5-10 additional papers on PPO implementation, curriculum learning in games, LSTM sequence prediction, and RL sample efficiency will be identified during literature review.